

SS 2 DIGITAL TECHNOLOGY WEEKS 1-6

Week: One

Definition of Word Processing

Word Processing is the use of computer software to create, edit, view, store, retrieve and print text documents. A text document on the other hand is a written communication like letters, reports, memos and so on. The software used for word processing is called a word processor.

Examples of word processors are:

1. Microsoft Word
2. WordStar
3. WordPerfect
4. word pro
5. Corel WordPerfect
6. Lotus notes
7. Perfect writer
8. Multimate
9. Notepad, e.t.c.

Advanced Word Processors tools include:

Watermark

A watermark is a faint background text or image, such as “DRAFT”, “CONFIDENTIAL” or a company logo, which appears lightly behind the main text of a document on every page. It is often used to indicate the status of a document (e.g. draft, sample, urgent) or to show ownership/authenticity.

To insert a watermark in Microsoft Word:

1. Click the “Design” tab on the Ribbon.
2. Click “Watermark” in the Page Layout tab.
3. Choose a built-in watermark (e.g. “DRAFT”, “CONFIDENTIAL”), or click “Custom Watermark” to type your own text or insert a picture.
4. Set the desired font, size, colour, layout (diagonal or horizontal) and transparency.
5. Click “OK”/“Apply” to insert the watermark on every page of the document.

Mail Merge

Mail merge is a feature in Microsoft Word that allows a user to combine a main document (such as a letter or certificate) with a data source (such as a list of names and addresses) in order to automatically produces multiple personalized copies of the same document.

Steps for setting up a mail merge:

1. Open a new or existing Word document to serve as the main document (e.g. an invitation letter).
2. Click the “Mailings” tab, then click “Start Mail Merge” and choose the type of document (e.g. Letters).
3. Click “Select Recipients” and choose a data source (e.g. “Use an Existing List” to select an Excel sheet, or “Type a New List”).
4. Click “Insert Merge Field” to place fields such as «FirstName» or «Address» into the document where personalized data should appear.
5. Click “Preview Results” to check how each merged document will look.
6. Click “Finish & Merge” and choose to print the documents or create individual merged documents/e-mails.

Macros

A macro is a set of recorded actions/commands (such as formatting steps) that can be saved and replayed automatically with a single click or shortcut key, in order to save time and ensure consistency when performing repetitive tasks in Word.

Steps for recording a macro:

1. Click the “View” tab, then click “Macros” and select “Record Macro”.
2. Type a suitable name for the macro and, if desired, assign it to a button or a keyboard shortcut.
3. Click “OK” to start recording, then perform the sequence of actions to be automated (e.g. formatting a heading).
4. Click “Macros” again and select “Stop Recording” once all the actions have been completed.
5. Run the saved macro at any time by clicking “Macros” → “View Macros” → selecting the macro → “Run”

Template

A template is a ready-made document layout with a predefined design, formatting, fonts and placeholder text, which serves as a starting point for creating new documents such as letters, resumes, memos, certificates and reports. Using a template saves time and ensures consistency and a professional appearance.

Steps for creating a document from a template:

1. Click “File” and then “New”.
2. Browse the available templates or search online templates using keywords (e.g. “resume”, “letter”, “certificate”).
3. Click on the desired template to preview it.
4. Click “Create” to open a new document based on the chosen template.
5. Replace the placeholder text and images with your own content, then save the document with a new file name.

Microsoft Word Table

A table in Word is an arrangement of data in rows and columns, useful for organizing information such as timetables, results and comparisons

Steps for inserting and adjusting a table:

1. Click the “Insert” tab, then click “Table”.
2. Drag across the grid to select the required number of rows and columns, or click “Insert Table” to type exact numbers.
3. Click inside the table to type data into the cells.
4. Use the “Table Design” tab to apply borders, shading and a table style.
5. Use the “Layout” tab to insert/delete rows and columns, merge or split cells, and adjust row height and column width.

Assignment

1. Design a simple certificate template and insert a “SAMPLE” watermark on it.
2. Explain, in about five sentences, how you would use mail merge to send a personalized admission letter to 5 new students.

Week: Two

Topic: Spreadsheet I

Functions and Formula in Microsoft Excel

A formula is a mathematical expression, beginning with an equal sign (=), that a user writes to perform a calculation using values or cell references, e.g. =B2+B3. A function is a predefined (built-in) formula in Excel that performs a specific type of calculation automatically, such as =SUM (B2:B5).

Commonly used functions in MS-Excel include:

- SUM — adds up a range of values, e.g. =SUM(B2:B5);
- AVERAGE — finds the mean of a range of values, e.g. =AVERAGE(B2:B5);
- MAX/MIN — find the highest/lowest value in a range;
- COUNT — counts the number of cells containing numeric data;
- IF — tests a condition and returns one result if true and another if false, e.g. =IF (B2>=50, “Pass”, “Fail”).

Steps for creating a PivotTable:

1. Select the range of data (including headings) to be summarized.
2. Click the “Insert” tab, then click “PivotTable”.
3. Choose where the PivotTable should be placed (a new worksheet or the existing one) and click “OK”.
4. In the PivotTable Fields pane, drag the required fields into the “Rows”, “Columns” and “Values” areas.
5. The PivotTable automatically summarizes the data (e.g. total sales per region) based on the chosen fields.
6. Adjust the summary type (Sum, Count, Average, etc.) by clicking on the field in the “Values” area and selecting “Value Field Settings”.

Chart Setting

A chart is a visual/graphical representation of worksheet data, such as a column chart, bar chart, line chart or pie chart, which makes it easier to compare, analyze and interpret data at a glance.

Steps for creating and adjusting a chart:

1. Select the range of data to be charted, including row and column labels.
2. Click the “Insert” tab, then click the “Charts” group and choose the desired chart type (e.g. Column, Bar, Pie, Line).
3. The chart is automatically inserted into the worksheet.
4. Use the “Chart Design” tab to change the chart style, colour, layout, and to add/edit the chart title, axis titles and data labels.
5. Use the “Format” tab to further adjust the chart's shapes, fonts and colours.
6. Click and drag the chart to reposition it, or drag its corner to resize it.

Assignment:

Explain, step by step, how you would create a PivotTable to show the total sales per region from a sales data sheet.

Week: Three (3)

Topic: Advanced Presentation Tools

Multimedia Integration

Multimedia integration is the process of combining different forms of media — such as text, images, audio, video and animation into a single presentation slide, in order to make the presentation more engaging, informative and interactive. This can only be achieved through the aid of a **Presentation package** which is the program used to display information usually in form of a slide show.

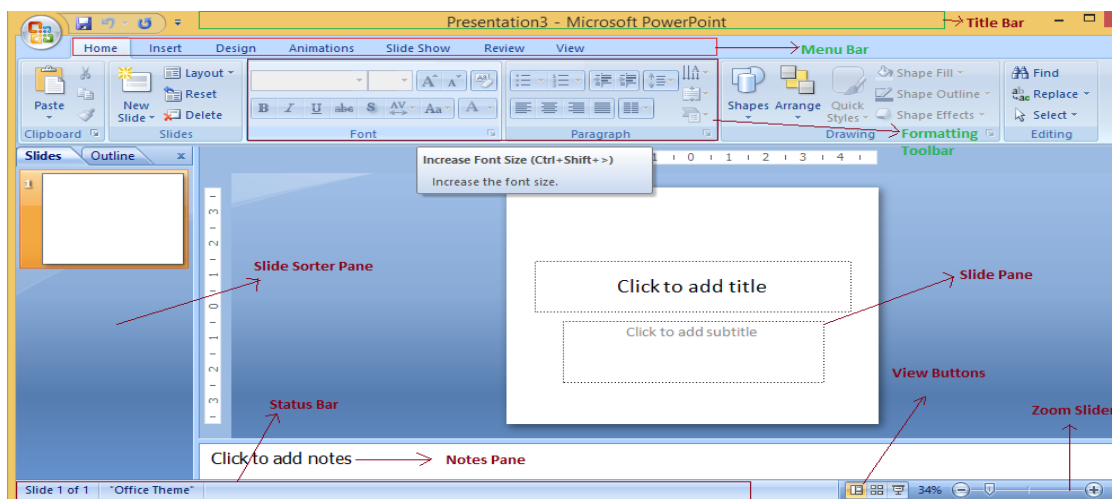
A **slide** is a single screen of a presentation. Just like every other application software that is built with a page setup, the presentation software labels its page as a slide; and more than one slide can be created in a presentation.

Examples of Presentation Software include:

- Emaze
- Google slides
- Macromedia Flash
- Microsoft PowerPoint
- Windows Movie maker
- Slide Wiki, e.t.c.

Steps for inserting multimedia elements into a slide:

1. To insert a picture: click “Insert” → “Pictures” → choose “This Device”, “Saved Images” or “Online Pictures” → select the image → click “Insert”.
2. To insert audio: click “Insert” → “Audio” → choose “Audio on My PC” or “Record Audio” → select/record the sound file.
3. To insert video: click “Insert” → “Video” → choose “This Device” or “Online Videos” → select the video file.



MICROSOFT POWERPOINT WINDOW

change the
background
ok.

Steps for applying a design setting:

1. Click the “Design” tab on the Ribbon.
2. Browse the gallery of built-in themes and click on one to apply it to all the slides.
3. Click “Variants” to change the colour scheme, fonts or background style of the chosen theme.
4. Click “Format Background” to set a custom colour, picture, gradient or texture as the slide background.
5. Click “Slide Size” to choose between Standard (4:3) and Widescreen (16:9) formats.
6. Click “Design Ideas” (where available) to let PowerPoint suggest professional layout designs for the current slide.

Assignment

Group Presentation using Ms-PowerPoint

Week: Four (4)

Topic: Memory Unit

Meaning of Memory

Memory refers to the component of a computer system used to store data, instructions and information either temporarily or permanently for processing. Computer memory is broadly classified (grouped) as follows:

- **Primary Memory** (Main Memory): is the memory that the CPU can access directly and quickly; it is further divided into:
 - RAM (Random Access Memory) — is a volatile memory that temporarily holds data and programs currently in use; its contents are lost when power is switched off.
 - ROM (Read Only Memory) — is a non-volatile memory that permanently stores essential start-up instructions (e.g. BIOS); its contents remain even when power is off and cannot ordinarily be altered by the user.
- **Secondary/Auxiliary Memory** (Storage): this is a non-volatile memory used to store data and programs permanently, even when the computer is switched off, e.g. hard disk drive (HDD), solid-state drive (SSD), flash drive, memory card, optical disc (CD/DVD).
- **Cache Memory**: a very fast, small-capacity memory located between the CPU and RAM, used to temporarily store frequently used data/instructions in order to speed up processing.
- **Registers**: A register is a small, high-speed storage location inside the CPU used for holding data, instructions, or memory addresses temporarily during processing operation. It is grouped into two: A special-purpose register which performs a dedicated, specific tracking function (e.g., holding instructions), and a general-purpose register which is flexibly utilized by programmers to store arbitrary data temporarily

Unit of Storage

The unit of storage is the standard measurement used to express the capacity/size of computer memory and storage devices. The basic units, from smallest to largest, are:

- **Bit** (Binary Digit): the smallest unit of data in a computer, represented as either 0 or 1;
- **Byte**: a group of 8 bits, usually representing a single character;
- **Kilobyte (KB)**: equal to 1,024 bytes;
- **Megabyte (MB)**: equal to 1,024 kilobytes;
- **Gigabyte (GB)**: equal to 1,024 megabytes;
- **Terabyte (TB)**: equal to 1,024 gigabytes
- **Petabyte (PB)**: equal to 1,024 terabytes

Conversion of Auxiliary Storage

Since each higher storage unit equals 1,024 units of the one immediately below it, aside from a byte which is equivalent to 8bits; values can be converted from one unit to another as follows:

- To convert from a smaller unit to a larger unit, divide by 1,024 (e.g. KB to MB, MB to GB)
- To convert from a larger unit to a smaller unit, multiply by 1,024 (e.g. GB to MB, MB to KB).

Examples:

1. Convert 4 Gigabytes (GB) to Megabytes (MB): $4 \text{ GB} \times 1,024 = 4,096 \text{ MB}$.
2. Convert 2,048 Megabytes (MB) to Gigabytes (GB): $2,048 \text{ MB} \div 1,024 = 2 \text{ GB}$.
3. Convert 3 Megabytes (MB) to Kilobytes (KB): $3 \text{ MB} \times 1,024 = 3,072 \text{ KB}$.
4. Convert 8,192 Kilobytes (KB) to Megabytes (MB): $8,192 \text{ KB} \div 1,024 = 8 \text{ MB}$.

Assignment

1. State three differences between primary memory and secondary (auxiliary) memory, giving two examples of each.
2. Give two clear differences between RAM and ROM.

Week: Five (5)

Topic: Logic Circuit I

Definition of Logic Circuit

A logic circuit is an electronic circuit made up of one or more logic gates that operate on binary inputs (0s and 1s) to produce a binary output, based on the principles of Boolean algebra. A logic gate is the basic building block of a logic circuit; it is an electronic device that performs a specific logical operation on one or more binary inputs to produce a single binary output.

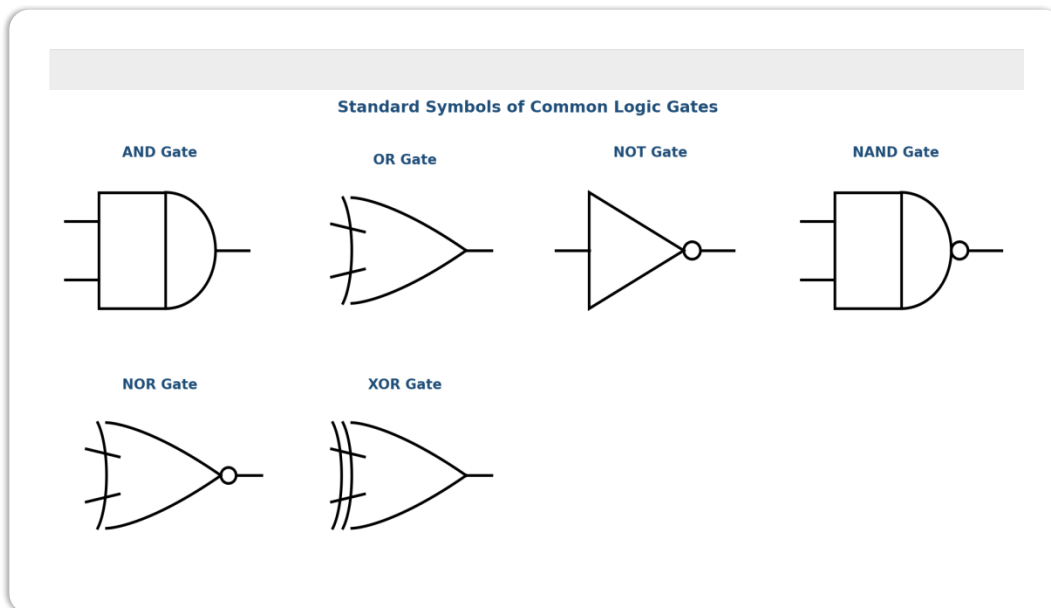
Types of Logic Gate

The common types of logic gate are:

- AND Gate — produces an output of high (1) only when all its inputs are 1
- OR Gate — produces an output of high (1) when at least one of its inputs is 1
- NOT Gate (Inverter) — produces an output that is the opposite (inverse) of its single input
- NAND Gate — the inverse (opposite) of the AND gate; it produces an output of 0 only when all its inputs are 1
- NOR Gate — the inverse (opposite) of the OR gate; it produces an output of 1 only when all its inputs are 0
- XOR Gate (Exclusive OR) — produces an output of 1 only when its inputs are different
- XNOR Gate (Exclusive NOR) — produces an output of 1 only when its inputs are the same.

Symbols of Logic Gate

Each logic gate has a unique standard symbol used to represent it in a logic circuit diagram, as shown below:



Truth Table

A truth table is a table that shows all the possible combinations of input values for a logic gate (or logic circuit) alongside the corresponding output produced for each combination. It is used to describe and verify the behaviour of a logic gate.

Truth Table for AND Gate

A	B	Output (A AND B)
0	0	0
0	1	0
1	0	0
1	1	1

Truth Table for OR Gate

A	B	Output (A OR B)
0	0	0
0	1	1
1	0	1
1	1	1

Truth Table for NOT Gate

A	Output (NOT A) / A'
0	1
1	0

Truth Table for NAND Gate

A	B	Output (A NAND B) / (A.B)'
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table for NOR Gate

A	B	Output (A NOR B) / (A+B)'
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table for XOR Gate

A	B	Output (A XOR B)
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0	0	0
0	1	1
1	0	1
1	1	0

Assignment

- (a) A logic circuit consists of four inputs (A, B, C, D) and one output (Z);
- (i) Identify the logic operation between A and B if it uses a NAND gate?
- (ii) Identify the logic operation between C and D if it uses an OR gate?
- (iii) State the complete Boolean logic equation for Z?
- (b) State the specific logic device used to invert an output signal?
- (c) Determine the output Z for the truth table row where A=1, B=1, C=0, D=0?

Week: Six (6)

Topic: System Development Cycle

Definition of System Development Cycle

The System Development Cycle (also called the System Development Life Cycle, SDLC) is a structured, step-by-step process followed by system analysts and programmers to plan, design, build, test, implement and maintain an information system, in order to solve a particular problem within an organization. It can also be defined as a project management technique that divides complex projects in a system into smaller, more manageable segments or phases.

Description of System Development Cycle

The System Development Cycle describes the logical sequence of activities that a new computer-based system passes through; from the moment a problem or need is identified to the point where the resulting system is fully installed, in use, and subsequently maintained. It ensures that a system is developed in an organized manner, reduces errors, and produces a system that truly meets the users' requirements.

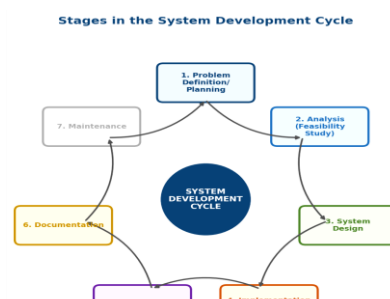
Stages in System Development Cycle

1. Problem Definition/Planning: in this phase, the existing problem or need is identified, and the objectives and scope of the new system are defined.
2. Analysis (Feasibility Study): here, the current system is studied in detail to understand its strengths and weaknesses, and a feasibility study is carried out to determine whether the proposed system is technically, economically and operationally possible.
3. System Design: in this phase, the structure of the new system is planned, including its input, output, files, database, users' interface and processing procedures.
4. Implementation (Coding): this is the system design is translated into actual computer programs using a suitable programming language.
5. Testing: the developed system/programs are tested using sample data to check for errors (bugs) and to confirm that the system works as expected.
6. Documentation: written records (user manuals, technical manuals) describing how the system works and how to use/maintain it are prepared.
7. Maintenance: the system is regularly reviewed and updated after installation to correct errors, improve performance, or adapt it to new requirements.

Diagram of System Development Cycle

The System Development Cycle is usually represented as a continuous cycle (as shown below), because system development does not simply end after implementation.

Note: — the maintenance stage often reveals new problems or needs, which restarts the cycle from the planning stage. This is why it is referred to as a “cycle” rather than a one-off, straight-line process.



Assignment

1. Briefly explain the primary function of at least five flowchart symbols?
2. Identify the structural meaning of five standard flowchart shapes?